Ex.No. 2: Implementing TDMA Slot Allocation in GSM.

Date:23-7-26

AIM: To implement TDMA (Time Division Multiple Access) slot allocation in GSM using
MATLAB and analyze system performance in terms of time slot duration, frame utilization,
and number of allocated slots.

Objective:

1. To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frame
time into equal time slots. This helps in understanding the time structure of GSM
communication.

2. To determine the Frame Utilization (%) based on the number of allocated time slots
compared to total available slots. This evaluates the efficiency of TDMA resource usage.

3. To compute the Number of Allocated Time Slots for multiple users in the GSM system.
This ensures proper scheduling and interference-free communication.

Procedure:

1. Define GSM frame parameters such as total frame duration and number of time slots.
2. Compute time slot duration by dividing frame duration by number of slots.
3. Assign time slots to multiple users based on availability.
4. Calculate frame utilization percentage using allocated slots vs total slots.
5. Determine total number of allocated slots for active users.
6. Plot time slot allocation and utilization using MATLAB graphs.
7. Display results in the command window for analysis..

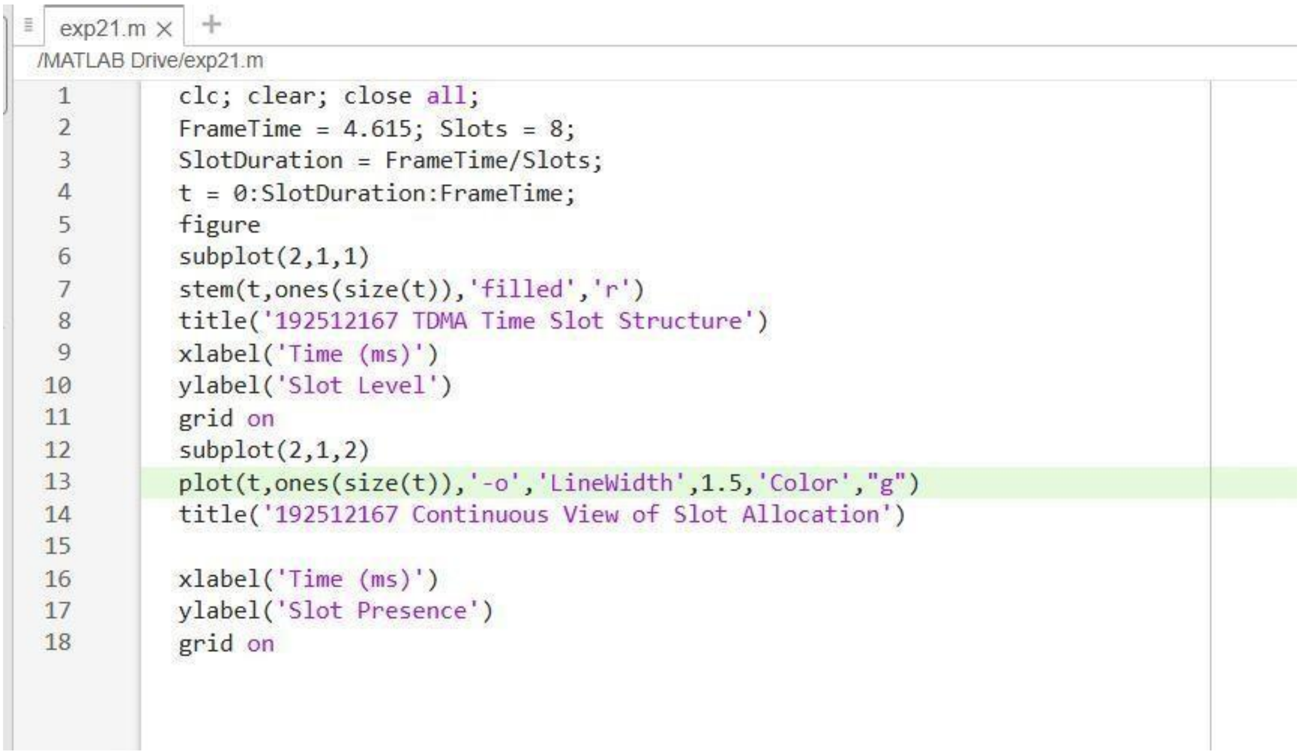
Objective 1: Matlab Code and Output clc;

clear; close all;

FrameTime = 4.615; Slots = 8;

SlotDuration = FrameTime/Slots; t =
0:SlotDuration:FrameTime; figure
subplot(2,1,1) stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level') grid
on subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5) title('Continuous
View of Slot Allocation')

xlabel('Time (ms)') ylabel('Slot Presence')
gridon

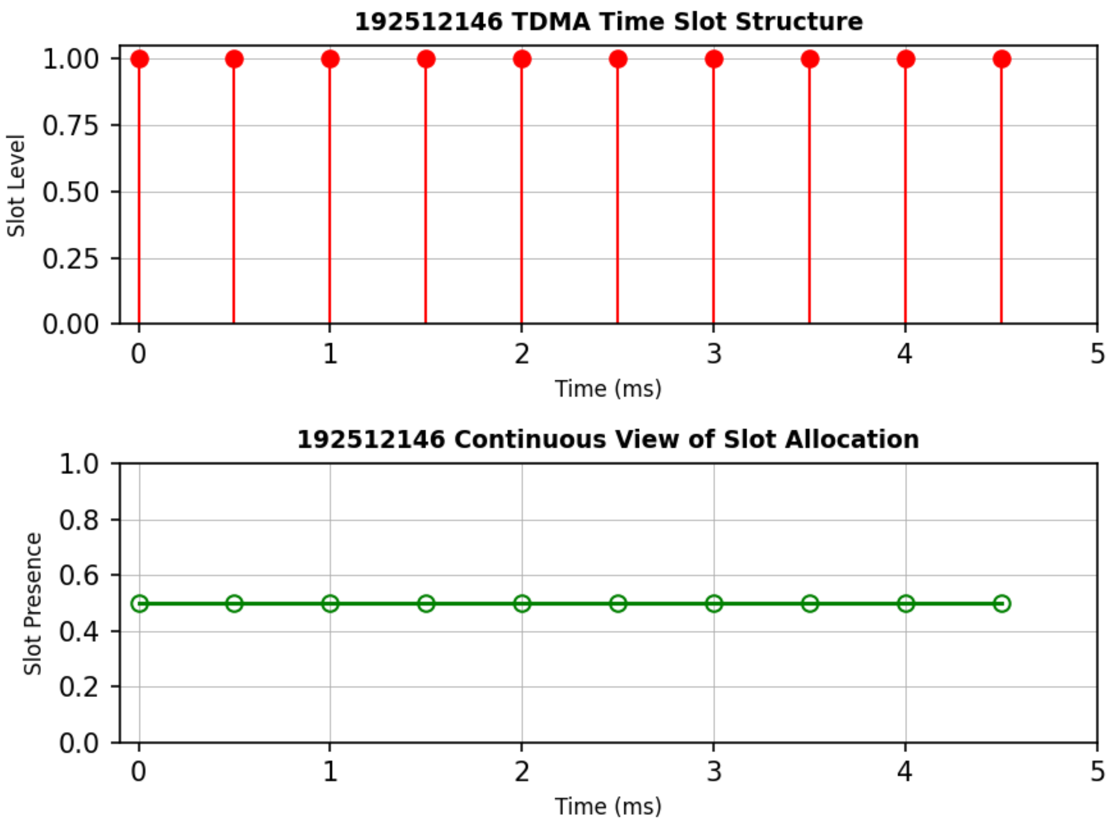


exp21.m

/MATLAB Drive/exp21.m

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled','r')
title('192512167 TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5,'Color',"g")
title('192512167 Continuous View of Slot Allocation')

xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```



Objective 2: Matlab Code and Output clc;
clear; close all;
TotalSlots = 8;
AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots; Utilization =
(AllocatedSlots/TotalSlots)*100; disp('Frame Utilization
(%)')
disp(Utilization) figure
% Graph 1: Simple Line Plot (SAFE - no bar function) subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2) title('GSM
Frame Utilization (Allocated vs Free)') xlabel('Slot Type Index
(1=Allocated, 2=Free)') ylabel('Number of Slots')
grid on

% Graph 2: Pie Chart (SAFE)
subplot(2,1,2) pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
legend('Allocated','Free')

exp21.m exp22.m

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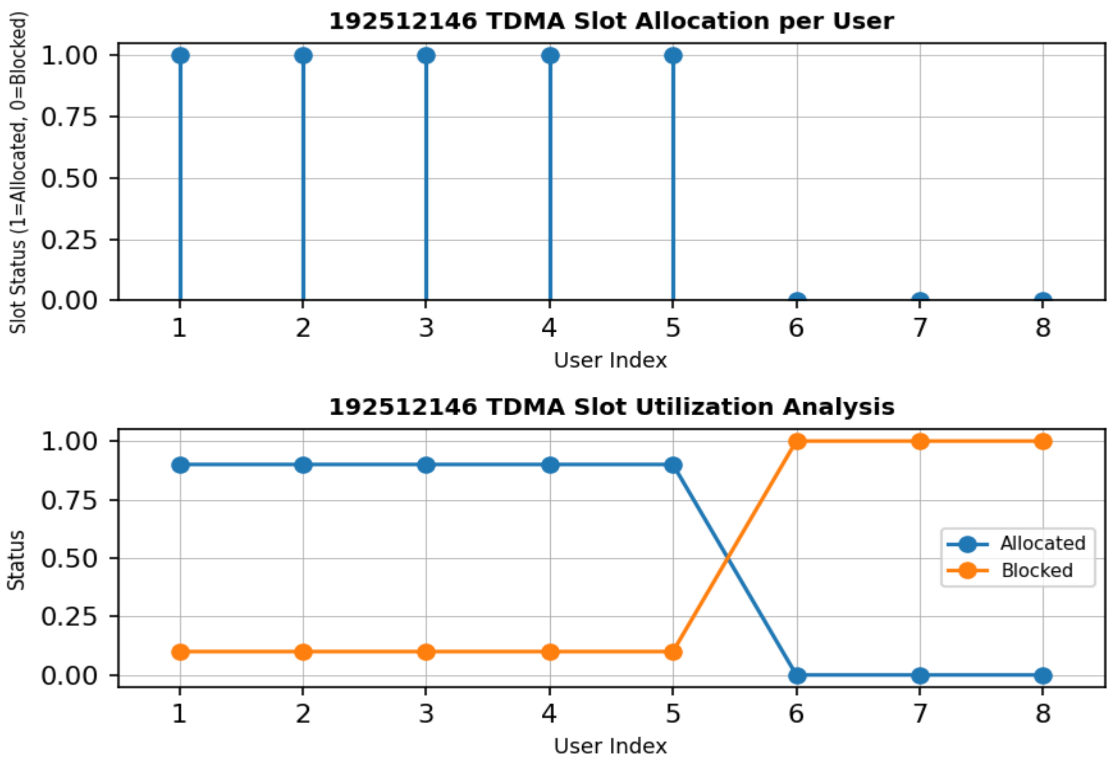
```
clc; clear; close all;
TotalSlots = 8;
AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots;
Utilization = (AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
disp(Utilization)
figure
% Graph 1: Simple Line Plot (SAFE - no bar function)
subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2,'Color','g')
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on

% Graph 2: Pie Chart (SAFE)
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
legend('Allocated','Free')
```



Command Window

```
     5          1
     6          0
     7          0
     8          0

>>
```

Objective 3: Matlab Code and Output clc;

clear; close all;
Users = 1:8; % Total users TotalSlots
= 5; % Available slots
Alloc = zeros(1,8); % Initialize allocation Alloc(1:TotalSlots) = 1; % Assign slots to first users
Blocked = 1 - Alloc; % Remaining users blocked disp('User Wise Slot Allocation')
disp(table(Users',Alloc','VariableNames',{'User','Status(1=Alloc ated,0=Blocked)'}))
figure
% Graph 1: Allocation per user
subplot(2,1,1)

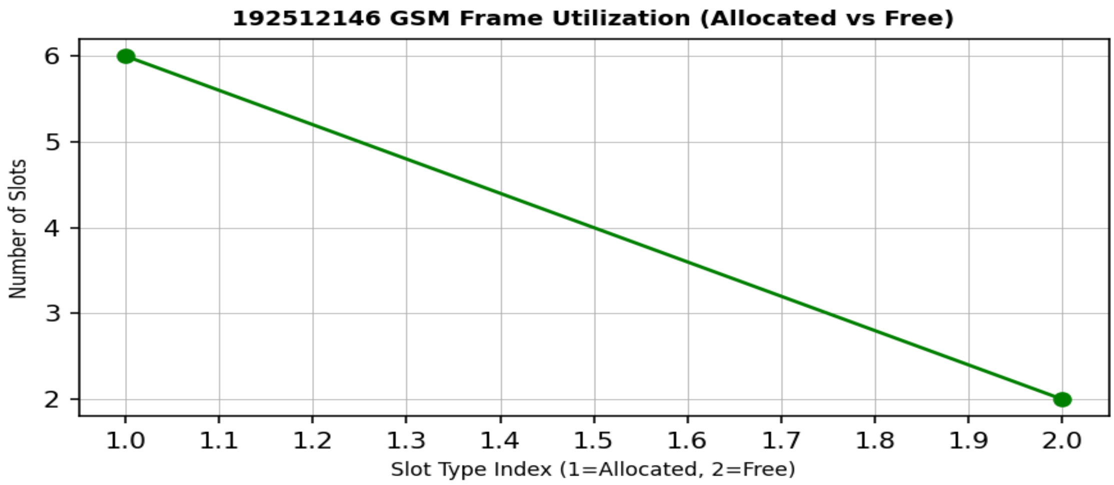
stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on
% Graph 2: System view (safe alternative to bar)
subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2) hold on
plot(Users,Blocked,'-s','LineWidth',2) title('TDMA Slot Utilization Analysis') xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked') grid on

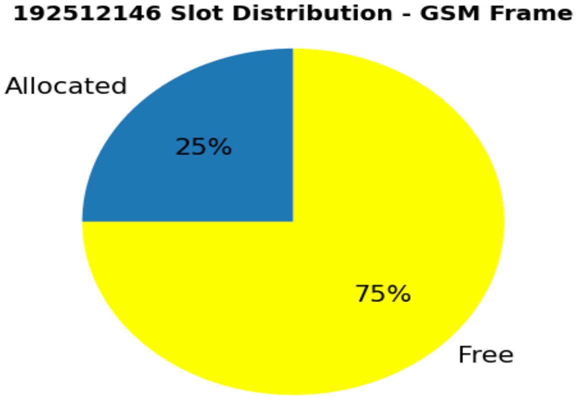
exp21.m × exp22.m × emf23.m ×

/MATLAB Drive/emf23.m

```
clc; clear; close all;
Users = 1:8; % Total users
TotalSlots = 5; % Available slots
Alloc = zeros(1,8); % Initialize allocation
Alloc(1:TotalSlots) = 1; % Assign slots to first users
Blocked = 1 - Alloc; % Remaining users blocked
disp('User Wise Slot Allocation')
disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))
figure
% Graph 1: Allocation per user
subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on
% Graph 2: System view (safe alternative to bar)
subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2)
```






```
Command Window
Frame Utilization (%)
    75

>>
```

Result:

1. The time slot duration in the GSM TDMA frame was successfully calculated, confirming equal division of the frame into fixed time intervals.

2. The frame utilization was effectively analyzed, showing the proportion of allocated and free slots in the TDMA system.

3. The slot allocation per user was successfully implemented, demonstrating that limited slots are assigned to users while excess users are blocked due to resource constraints.